

Computational fluid dynamics analysis of pollutant dispersion around a high-rise building: Impact of surrounding buildings

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ARTICLE INFO

Keywords:

Computational fluid dynamics (CFD)
Large-Eddy Simulation (LES)
Pollutant dispersion
High-rise building
Surrounding buildings

ABSTRACT

Understanding the detailed mechanism of pollutant dispersion in urban areas is a critical aspect of air quality control in residential areas. In this study, wind tunnel measurements and high-resolution computational fluid dynamic (CFD) based on large eddy simulation (LES) were used for the detailed analysis of the pollutant dispersion around a high-rise building for an isolated case and also a case when the central high-rise building was surrounded by several low-rise buildings. The pollutant was released on the leeward façade of the high-rise building at the pedestrian height. It was shown that the presence of the surrounding buildings significantly alters the pollutant transfer mechanism around the high-rise building. In the surrounded case, the pollutant in the wake region shifted vertically toward the roof of the high-rise building. As a result, at a location of 2/3 of the high-rise building height, the mean concentration is five times larger than the isolated case, which means the residents on higher floors experience more pollution concentration in the case of the surrounded case even though they are located much higher than the canyon rooftop and the pollutant source. In both cases, the contribution of the turbulent diffusive flux in the streamwise direction is comparable with the mean convective flux while they have opposite directions. Detailed analyses of the pollutant mass transfer rate budget around different control volumes around the high-rise buildings highlighted the effect of the surrounding buildings. It is concluded that there is a risk of underestimating pollutant concentrations around a high-rise building, especially near the upper floors if the surrounding buildings are not properly considered.

1. Introduction

Air pollution in urban areas is one of the serious social problems in large cities. According to the World Health Organization (WHO)'s publication on Global Health Risk (2009), urban outdoor air pollution was ranked as the 14th global risk factor for mortality in 2004 [1]. In order to maintain the air quality in the urban residential area, it is necessary to properly remove and/or dilute the pollutants generated inside the urban areas. Needless to say, since the pollutant dispersion is strongly affected by the wind, accurate prediction of pollutant dispersion under complex wind field formed in the built-up environment has been an important research topic. Numerous studies have been conducted on the prediction of pollutant dispersion in the built-up environment, which has been reviewed in Refs. [2–14]. As pointed out in those reviews, the computational fluid dynamics (CFD) simulation has several advantages over physical measurements and empirical statistical models, namely i) it can provide whole flow and dispersion field data

and be performed at full scale, ii) it does not suffer from similarity requirements, therefore it has been recognized as a powerful tool for pollution dispersion problems [15–21].

Previous studies of CFD modeling on pollutant dispersion around buildings have mainly focused on two generic and idealized configurations: an isolated building and an isolated street canyon [15]. In these studies, by ignoring or simplifying the complex interactions among the buildings that formed in actual urban configurations, it has been possible to elucidate the detailed mechanisms of turbulent pollutant transport and evaluate the performance of CFD modelling, e.g. Refs. [22–24]. While both configurations are derived from strong simplifications of reality, the flow and dispersion processes include complex interactions among the buildings in the urban environment. Arrays of buildings or building blocks have been considered as another generic configuration, e.g. Refs. [25–27].

On the other hand, there are very few studies focusing on the influence of a high-rise building on pollutant dispersion in neighborhood

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<https://doi.org/10.1016/j.buildenv.2023.110895>

Received 2 August 2023; Received in revised form 29 September 2023; Accepted 2 October 2023

Available online 4 October 2023

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areas, although such situations are very common in big cities. The impact of a high-rise building on the dispersion field in urban areas surrounded by adjacent buildings is expected to be significant, because it includes the complex interaction between the atmospheric boundary layer (ABL) and flows around a high-rise building such as a downwash corner stream, etc. Furthermore, such configurations include street intersections that play an important role in determining pollutant concentrations in the urban canopy because vehicle emissions often increase in the vicinity of road intersections, and the complex flow patterns that occur within the intersection determine the pollutant fluxes into adjoining streets and into the atmosphere [28]. On the other hand, although some case studies of pollutant dispersion for real urban configurations have been reported [29–31], irregular building geometries make it difficult to investigate detailed pollutant dispersion mechanisms in the urban canopy affected by a high-rise building.

Brixey et al. [32] investigated the effect of a tall tower on flow and dispersion through a model urban neighborhood based on wind tunnel and CFD simulations. They found that the tower had three significant effects on pollutant dispersion in the surrounding street canyons: drawing the plume laterally towards the tower, greatly enhancing the vertical dispersion of the plume in the wake of the tower, and significantly decreasing the residence time of pollutants in the wake of the tower. Zhang et al. [33] studied the pollutant diffusion around a typical Hong Kong high-rise building model by CFD. Their results indicate that the pollutant migrated downwards due to the downwash for the case with windward pollutant emission. In contrast, in the case of leeward position emission, dispersion is dominated by intense turbulent mixing in the near wake and characterized by the upwind migration of the pollutant. Aristodemou et al. [34] carried out detailed CFD and wind tunnel studies to assess the effect of high-rise buildings on the dispersion of air pollution in their vicinity. Their results clearly showed that the high-rise building significantly affected the surrounding air flows and dispersion patterns, with the generation of “dead-zones” and high-concentration “hotspots”. As seen in these results, high-rise buildings have a significant impact on the surrounding pollutant dispersion. However, because these studies were conducted for specific building configurations, the detailed mechanism of pollutant dispersion is not fully understandable. Therefore, more detailed investigations on the surrounding dispersions are needed with more generic configurations including the interaction between the neighborhood buildings.

A generic building group including a high-rise building with a 1:1:3 shape was adopted as a target configuration for this study. Pollutant was released from the bottom of the leeward surface of the high-rise building. Such a moderate level of abstraction can be a good representative of the flow and dispersion fields in urban environments. This also means linking the knowledge of the isolated building, the isolated street canyon, and the building array in previous studies. To the authors' knowledge, there are few examples of such studies. One of the problems in applying CFD to such dispersion field is that there is no experimental database that can be used for validation of the computational model. First, we conducted detailed wind tunnel experiments to collect measurement data on wind velocity and concentration distributions with the intention of performing such a validation study. The CFD method used is validated using the results of the wind tunnel experiments. In the previous study [35], we compared the prediction accuracy of mean and fluctuating wind velocities by steady Reynolds averaged Navier-Stokes equation (RANS) models and large eddy simulations (LES) for the same configurations. We found that RANS models have generally high prediction accuracy for high-wind velocity regions, but there are inherent shortcomings in their prediction accuracy for low-velocity regions. Because such shortcomings could be a serious problem in the prediction of pollutant dispersion field, we use the LES in this study. Using the high-resolution information of flow and concentration fields obtained by the carefully validated LES, the impact of the high-rise building on the surrounding pollutant dispersion is discussed.

This paper consists of the following sections: Section 2 describes the

details of the experimental setup to create the experimental database; Section 3 describes the setup of CFD simulations; Section 4 reports the mesh sensitivity check and validation for the velocity and concentration fields. In Section 5, the impacts of surrounding buildings on the transport mechanism of the pollutant concentration are discussed. Section 6 summarizes the results of this paper.

2. Experimental setup

2.1. Case description

Two different cases were considered with the same high-rise building in the center which had a height of $3H$, with $H = 0.1\text{ m}$ as the reference height. In the first case, namely the isolated case, there were no surrounding buildings. In the second case, namely, the surrounded case, there were eight low-rise buildings with a height of H around the central high-rise building that were arranged uniformly with a streamwise and lateral distance of H , as depicted in Fig. 1. All the buildings had the same cross-section with a dimension of $0.1\text{ m} \times 0.1\text{ m}$. The pollutant was released through a small hole on the leeward of the high-rise building at a height of 0.01 m and had a dimension of $0.005\text{ m} \times 0.005\text{ m}$. The wind direction is perpendicular to the windward façade and was kept constant for all the cases. The coordinate origin was set at the center point of the central building on the ground surface.

In our previous study [36], we showed that the flow field around a building surrounded by one layer of surrounding buildings is almost identical to a case of continuous cube arrays with more than 7 layers of surrounding buildings. Hence, the flow and dispersion fields in the street canyon cavities are considered to be less sensitive to the number of the surrounding building layers, and the one layer of surrounding buildings, as considered in this study, can represent the urban flow condition.

2.2. Wind tunnel experimental setup

The current building configuration was used in the author's previous studies for the analyses of the pedestrian wind environment and turbulent flow around the building using high-resolution numerical simulation and wind tunnel experiments [35–37]. The same wind tunnel setup was used in the current study. All the measurements were conducted in the atmospheric boundary layer wind tunnel at Niigata Institute of Technology. The wind tunnel has a length of 13 m and the test section dimension is $1.8\text{ m} \times 1.8\text{ m}$ [38–42]. The inside view of the wind tunnel is shown in Fig. 2(a) for the pollution concentration measurement for the surrounded case. The incident vertical profiles of the time-averaged streamwise velocity U_H and the turbulent kinetic energy (TKE), k is shown in Fig. 2(b) which were measured at the center of the empty turntable. These values were normalized by the reference velocity of $U_H = 2.8\frac{\text{m}}{\text{s}}$ which was measured at the reference height H in the centre of the empty turntable. TKE was obtained as $k = (\overline{u'^2} + \overline{v'^2} + \overline{w'^2})/2$ by measuring the three components of the Reynolds stress ($\overline{u'^2}$, $\overline{v'^2}$, $\overline{w'^2}$) in the streamwise (x), lateral (y), and vertical (z) directions. The fluctuating component u'_i of the instantaneous velocity u_i was calculated as $u'_i = u_i - U_i$ with U_i denoting the time-averaged velocity component i .

The Reynolds number based on H and U_H was 1.9×10^4 and the aerodynamic roughness length was $z_0 = 0.0002\text{ m}$, which was obtained by curve fitting of the incident flow profile. More details about the wind tunnel characteristics and incident flow parameters can be found in Refs. [35–37].

Ethylene (C_2H_4) with a concentration of 5% wt (C_e) was used as the pollution gas at a constant flow rate of $Q_e = 0.2\text{ L/min}$ through the releasing point on the leeward façade of the central high-rise building. The reference concentration for normalization was calculated as $C_0 = \frac{C_e Q_e}{H^2 U_H} = 5.95\text{ ppm}$.

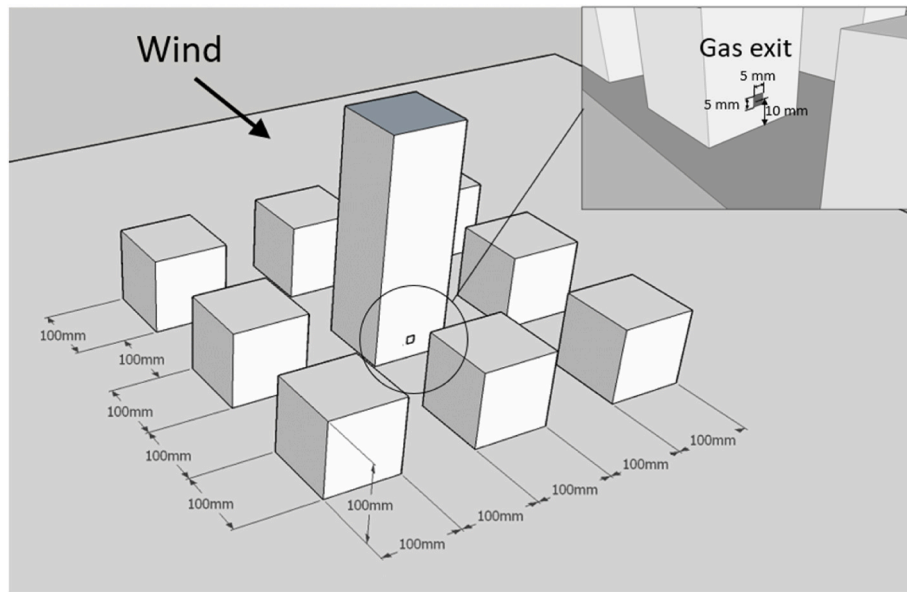


Fig. 1. Building configuration (surrounded case).

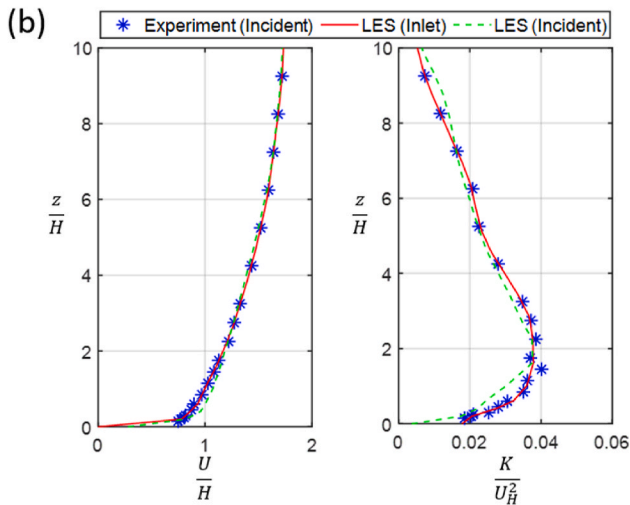
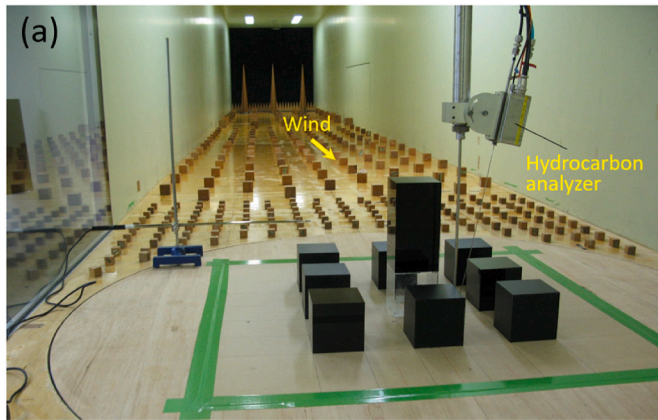


Fig. 2. (a) Inside view of the wind tunnel for the surrounded case. (b) Vertical profiles of approaching flow.

2.3. Measurement techniques

A split fiber probe (Dantec Dynamics 55R55) and a constant temperature anemometer module (Dantec Dynamics 90C10) were used to measure the instantaneous velocity components velocity u , v , and w , in the streamwise (x), lateral (y), and vertical (z) directions by rotating the probe head in the corresponding direction. The sampling frequency was 1,000 Hz with a sampling time of 60 s. The instantaneous pollution concentration c was measured by a high-speed total hydrocarbon analyzer (Technica, HTHCA-01) at a sampling frequency of 1000 Hz for a sampling time of 120 s. All the measurements were repeated several times to measure the relative expanded uncertainty which was estimated to be less than 10% [43].

3. CFD simulation setup

3.1. Computational domain and mesh

The computational domain is shown in Fig. 3(a) which was made based on the AIJ guidelines [44] and had a dimension of $1.8\text{ m} \times 1.8\text{ m} \times 5.6\text{ m}$. The cut-cell method was used for the mesh generation by ANSYS Mesh which was proven to be an efficient mesh generation technique for simple generic buildings [35,45] and buildings with complex geometry [46,47]. The mesh setting was obtained after a mesh sensitivity analysis for three different mesh settings for coarse, medium, and fine meshes, named Mesh A, Mesh B, and Mesh C, respectively. To reduce the computational cost, the cell element size was increased uniformly from the fetch region toward the central building region. Details of the cell element size are shown in Table 1. The numbers of cell elements for Meshes A, B, and C, were 1,327,676, 2,507,540, and 3,202,910, respectively for the surrounded case.

The boundary layer mesh over the ground surface upstream of the building region had a first layer height of 0.0025 m ($=0.025\text{ H}$) and an expansion ratio of 1.08, resulting in an average $y^+ = 28$ which is required to satisfy the condition for the logarithmic velocity profile for the rough boundary layer over the ground surface [48]. Furthermore, more than four mesh layers were created below the pedestrian height for reliable prediction of the pedestrian wind environment [44]. For the isolated building case, the same computational domain was used while the mesh setting was kept very close to the surrounded case that was described above. With the cell element sizes as Mesh B (see Table 1), the

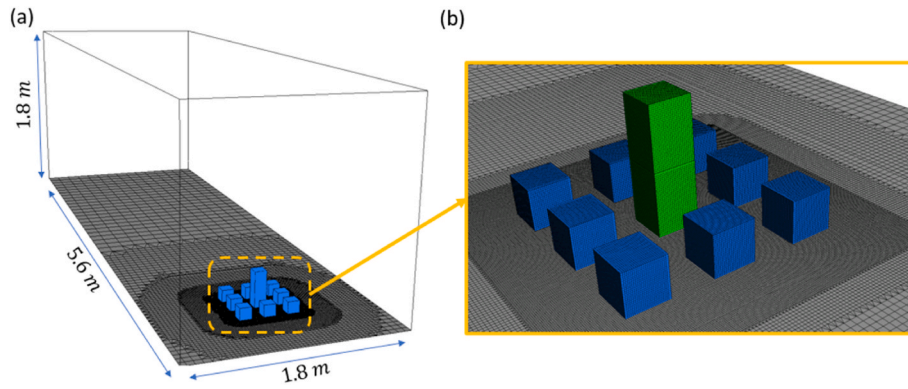


Fig. 3. (a) Computational domain and (b) grid cell distribution over the building surfaces.

Table 1

Cell element size for mesh sensitivity study.

Element Size [m]	Mesh A	Mesh B	Mesh C
Central building	0.0025	0.0025	0.0018
Surrounding building	0.005	0.0025	0.0018
Area around the buildings	0.025	0.025	0.0018
Area far from the buildings	0.06	0.032	0.015
No. elements	1,327,676	2,507,540	3,202,910

number of cell elements was 781,826 for the isolated case. The same boundary layer mesh configuration was also applied for the isolated case resulting in an average $y^+ = 26$ over the ground surface.

3.2. Boundary conditions

The vertical profile of the time-averaged streamwise velocity U in the wind tunnel, which followed a power-law profile $\frac{U}{H} = \left(\frac{z}{H}\right)^p$ with an exponent of $p = 0.25$, was directly used as the inlet boundary condition for the time-averaged U , as shown in Fig. 2(b). For the generation of the velocity fluctuations in the inlet boundary plane in LES, the Vortex method [49] was used. The number of vortices (N) was determined as $N = \frac{N_{cell-face}}{4} = 1336$ with $N_{cell-face}$ as the number of cell faces on the inlet boundary plane. However, $N = 1000$ was adopted, because this is the maximum value of N in ANSYS Fluent, which is the commercial CFD solver used in this study. In this method, a parameter that controls the size of a vortex particle is calculated from given values of TKE and the dissipation rate of the TKE (ϵ). The profile of TKE was given by the measured one in the wind tunnel, which is also depicted in Fig. 2(b), and that of ϵ was calculated as follows [44]:

$$\epsilon(z) = \sqrt{0.09k(z)} \frac{U_H}{H} P \left(\frac{z}{H} \right)^{p-1} \quad (1)$$

For the ground surface and the building walls, the smooth wall boundary condition was applied because the buildings were made of plastic materials with smooth surfaces and the turntable in the test section of the wind tunnel was made of smooth wooden materials. Smooth wall boundary condition was used in previous similar studies for the building and ground surfaces [29,35,45]. The horizontal homogeneity of the inlet profile was verified by conducting an empty-domain test case with the same CFD setting and mesh distribution over the ground as the case with buildings. The vertical profiles of U and TKE in the building's location in the empty domain (incident profile) are shown in Fig. 2(b). The inlet and incident profiles show small deviations which verifies the horizontal homogeneity of the inlet profile.

For the side and top surfaces, a symmetric boundary condition was considered while a constant static pressure was implemented to the outlet boundary plane. An inlet velocity boundary condition with a uniform velocity of $0.133 \frac{m}{s}$, corresponding to the setting emission flow

rate of the experiment, was considered for the pollutant source exit while no turbulence was considered.

3.3. CFD solver settings

The governing equations for the mass, momentum, energy, and species transfer were discretized by the finite volume approach and solved using ANSYS Fluent 19.2. The turbulence modeling by LES was performed using the standard Smagorinsky model with a constant of $C_s = 0.12$, which was shown to be appropriate for the turbulent flow around buildings [45,50–56]. Ethylene was considered a non-reactant (passive) species that has a molecular weight close to the air, so a constant density was assumed for the air-ethylene mixture. Hence, a passive scalar transport equation with an SGS Schmidt number of 0.7 was solved for the dispersion of ethylene in the air. As will be shown in the following, the SGS contribution to the total TKE was small hence it was not considered in the mass flux calculations of the pollutant when only the contribution of the resolved part was considered. The pressure-implicit with the splitting of operators (PISO) solver was used for the coupling of momentum and continuity equations while the bounded central difference and second-order scheme were used for the spatial discretization of the momentum equation and pressure interpolations. The second-order implicit scheme was used for the transient formulations. A constant time-step of $\Delta t = 0.0005$ s was considered to satisfy the Courant number condition ($Co < 1$). The data acquisition duration for the statistical calculations was 45 s. The statistical convergence was checked by monitoring the development of the time-averaged velocity and concentration at several locations inside the street canyon. The same solver setting and sampling time were considered for the isolated and surrounded cases.

4. Mesh sensitivity and validation studies

4.1. Mesh sensitivity check

A detailed mesh sensitivity study was conducted in the previous study for the same building configuration and airflow conditions [35]; hence, the results of the mesh sensitivity are briefly explained here for the surrounded case. The time-averaged velocity magnitude over a vertical plane ($\frac{z}{H} = 0$) and a horizontal plane ($\frac{z}{H} = 0.1$) behind the central high-rise building are shown for Mesh A, B, and C in Fig. 4(a). The highest deviation was observed between the results by Mesh A and Mesh C, corresponding to the fine and coarse meshes with a root mean square error (RMSE) of 0.035. The majority of deviations were found at low-velocity regions with absolute errors larger than 10%. The prediction results by Mesh B (medium mesh setting) are very close to the results by Mesh C with RMSE = 0.018, indicating the low sensitivity of the mesh setting for Mesh B. The absolute error at low-velocity regions by Mesh B was less than 10%.

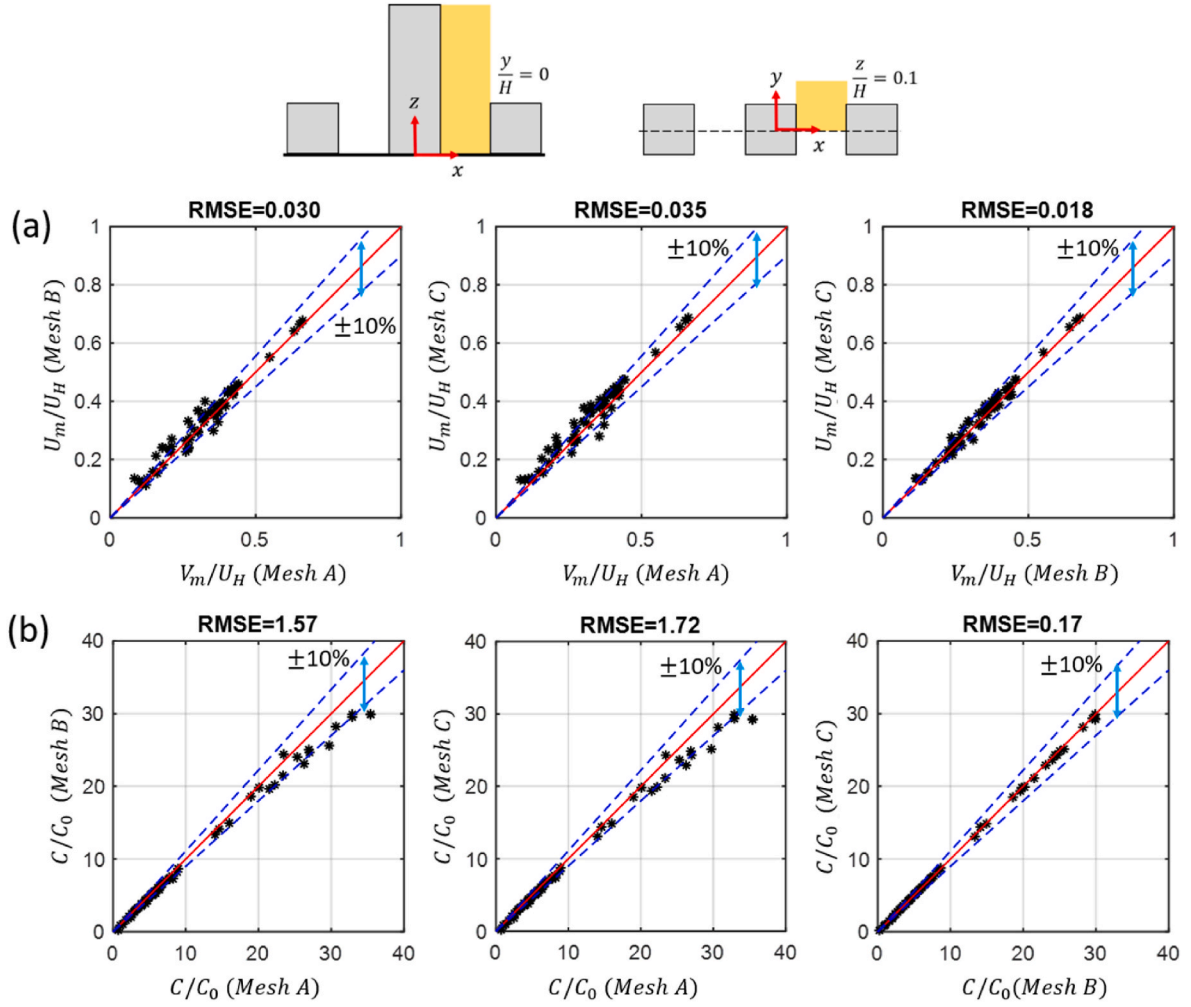


Fig. 4. Comparison of (a) velocity magnitude and (b) TKE for different mesh settings for the surrounded case.

The normalized time-averaged concentration $C^* = C/C_0$ is also compared for different mesh settings, as shown in Fig. 4(b). Generally, Mesh A predicted the concentrations field in good agreement with Mesh B and C with few points with absolute deviations of more than 10% at high-concentration values near the pollutant source. The deviation between the concentration prediction by the medium and fine mesh is very close with an RMSE = 0.17, which shows the appropriateness of Mesh B regarding the mesh sensitivity.

Moreover, the quality of index (IQ) was estimated over the same vertical and horizontal planes to make sure about the adequacy of the mesh resolution for Mesh B [57]. According to Ref. [57], IQ is defined as the ratio of the resolved kinetic energy to the total kinetic energy and should be greater than 80% in critical areas for LES [58,59]. The average IQ was approximately 90% over the measurement planes which was higher than the threshold of 80% and showed that the majority of TKE was resolved in LES with Mesh B. This also highlights the small contribution of SGS turbulence to the total kinetic energy and consequently makes it possible to ignore the SGS contribution in the flux calculation of pollution. More details about the mesh sensitivity study can be found in the previous paper by the authors [35]. For the rest of the calculations, Mesh B was used.

4.2. Validation of velocity field

The coordinates of the measurement points on vertical and horizontal planes in the downstream cavity behind the central high-rise building are shown in Fig. 5(a). The height of the horizontal plane

was $\frac{z}{H} = 0.1$ which is close to the pedestrian-level height that is used for wind velocity and pollution evaluation. The scatter plots of U and TKE over the vertical and horizontal plane are compared for LES and experimental results, as depicted in Fig. 5(b) and (c), respectively. The dashed lines indicate absolute deviations of 10% and 20% between the experimental and numerical results. As shown in Fig. 5(b), the velocity prediction by LES is in close agreement with the experiment as most points in the scatter plot are within the 10 and 20% deviation range. The largest deviations were observed in the low-velocity region in the wake region behind the high-rise building ($\frac{U}{U_H} \leq -0.5$) which may be contributed to the uncertainty of the experimental measurement technique and numerical uncertainties of LES. As can be seen in Fig. 5(c), the prediction accuracy of LES for TKE is reasonable for the majority of the measurement points with deviations of less than 20%, however, the deviations with experimental results were higher than that observed for the U , specifically at the locations with low TKE in the range of $0.025 \leq k/U_H^2 \leq 0.06$. The possible reason for such higher deviations in this range, in addition to the numerical uncertainties, is the measurement uncertainties due to the sensor positioning in the wake region. The same tendency by LES was also reported in Ref. [60] for an idealized street canyon flow. As explained in Section 2.3, the three components of the fluctuating velocity were measured by rotating the probe in the corresponding direction, which can possibly add more uncertainties for the calculation of TKE which is a summation of the three fluctuating components.

The validation metrics, including (FAC2), hit rate (q), fractional bias

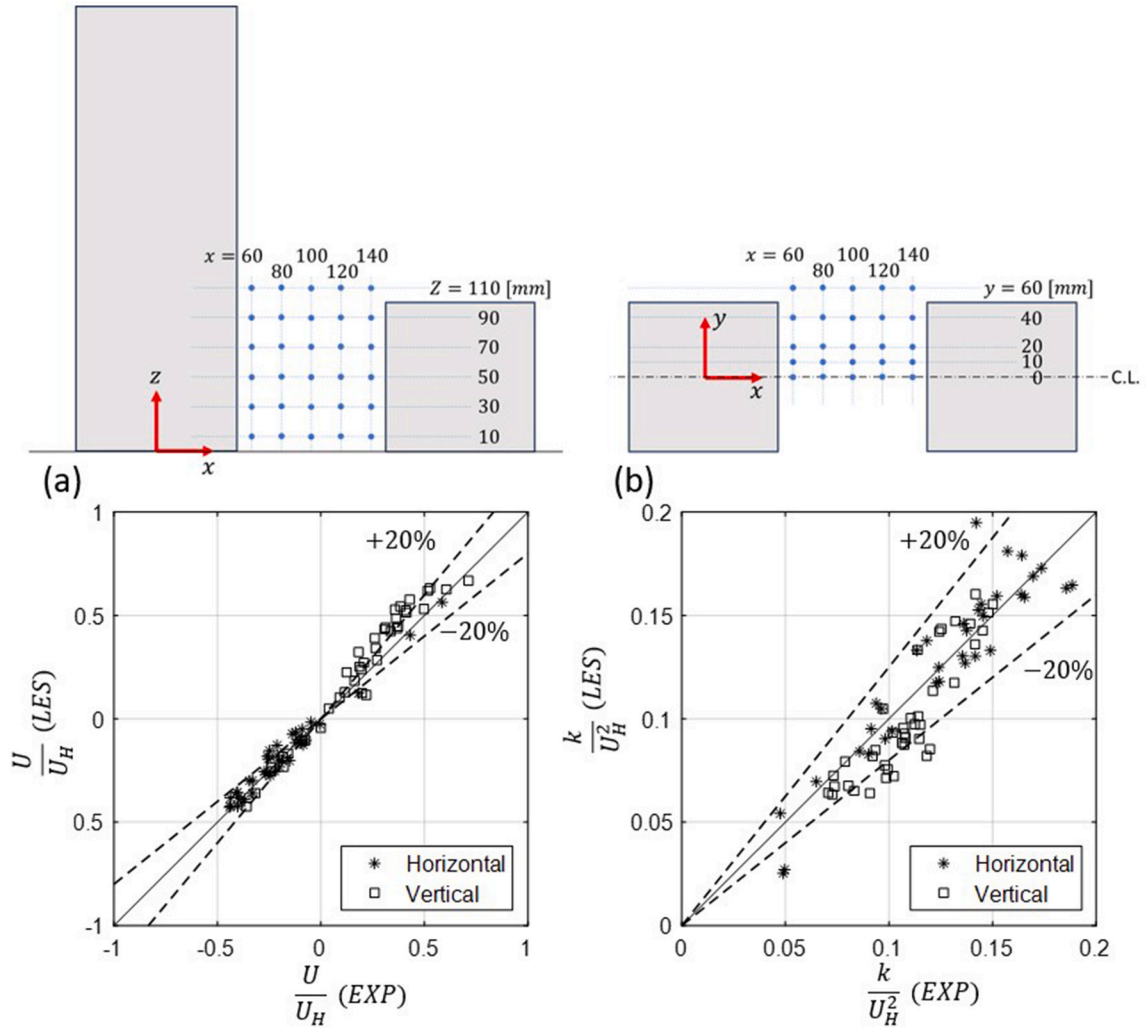


Fig. 5. (a) Coordinate of the measurement points in the wind tunnel experiments. Scatter plot of (b) the time-averaged streamwise velocity U and (c) TKE by LES and the experiment for the surrounded case.

(FB), and normalized mean square error (NMSE) were calculated for the surrounded case for the qualitative assessment of LES in the prediction of the normalized streamwise velocity ($\frac{U}{U_H}$) and TKE ($\frac{k}{U_H^2}$):

$$FAC2 = \frac{1}{n} \sum_{i=1}^n N_i$$

$$N_i = \begin{cases} 1 & \text{for } 0.5 \leq \frac{P_i}{Q_i} \leq 2 \\ 1 & \text{for } |Q_i| \leq W \text{ and } |P_i| \leq W \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

$$q = \frac{1}{n} \sum_{i=1}^n N_i \quad \text{with } N_i = \begin{cases} 1 & \text{for } \left| \frac{P_i - Q_i}{Q_i} \right| \leq D \\ 1 & \text{for } |Q_i - P_i| \leq W \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

$$FB = \frac{\langle Q \rangle - \langle P \rangle}{0.5(\langle Q \rangle + \langle P \rangle)} \quad (4)$$

$$NMSE = \frac{\langle (P - Q)^2 \rangle}{\langle Q \rangle \langle P \rangle} \quad (5)$$

where Q_i and P_i are the observed (measured) and predicted (computed)

values of a given variable, respectively, and n is the number of data points. D is set to 0.25, whereas W is set to 0.05 and 0.005 for the streamwise velocity and TKE, respectively [61,62]. According to Refs. [63,64], the acceptable criteria for a successful validation are $FAC2 \geq 0.5$, $q \geq 0.66$, $|FB| \leq 0.3$, and $NMSE \leq 1.5$. The high prediction accuracy of LES was confirmed by obtaining all the validation metrics in the acceptable range, as shown in Table 2. For U/U_H , hit rate and FAC2 was 0.70 and 1.00, respectively, while for TKE, hit rate, FAC2, FB, and NMSE were respectively 0.88, 1.00, -0.05, and 0.04.

4.3. Validation of concentration field

The scatter plot of the normalized time-averaged concentration ($C^* = \frac{C}{C_0}$) for all the measurement points over the horizontal ($\frac{z}{H} = 0.1$) and vertical ($\frac{y}{H} = 0$) planes is shown in Fig. 6(a) for the surrounded case. The dashed lines representing the absolute deviations of 10% and 20%

Table 2

Validation metrics for the time-averaged streamwise velocity and TKE for the surrounded case.

	Hit rate (q)	FAC2	FB	NMSE
U	0.70	1.00	–	–
TKE	0.88	1.00	–0.05	0.04

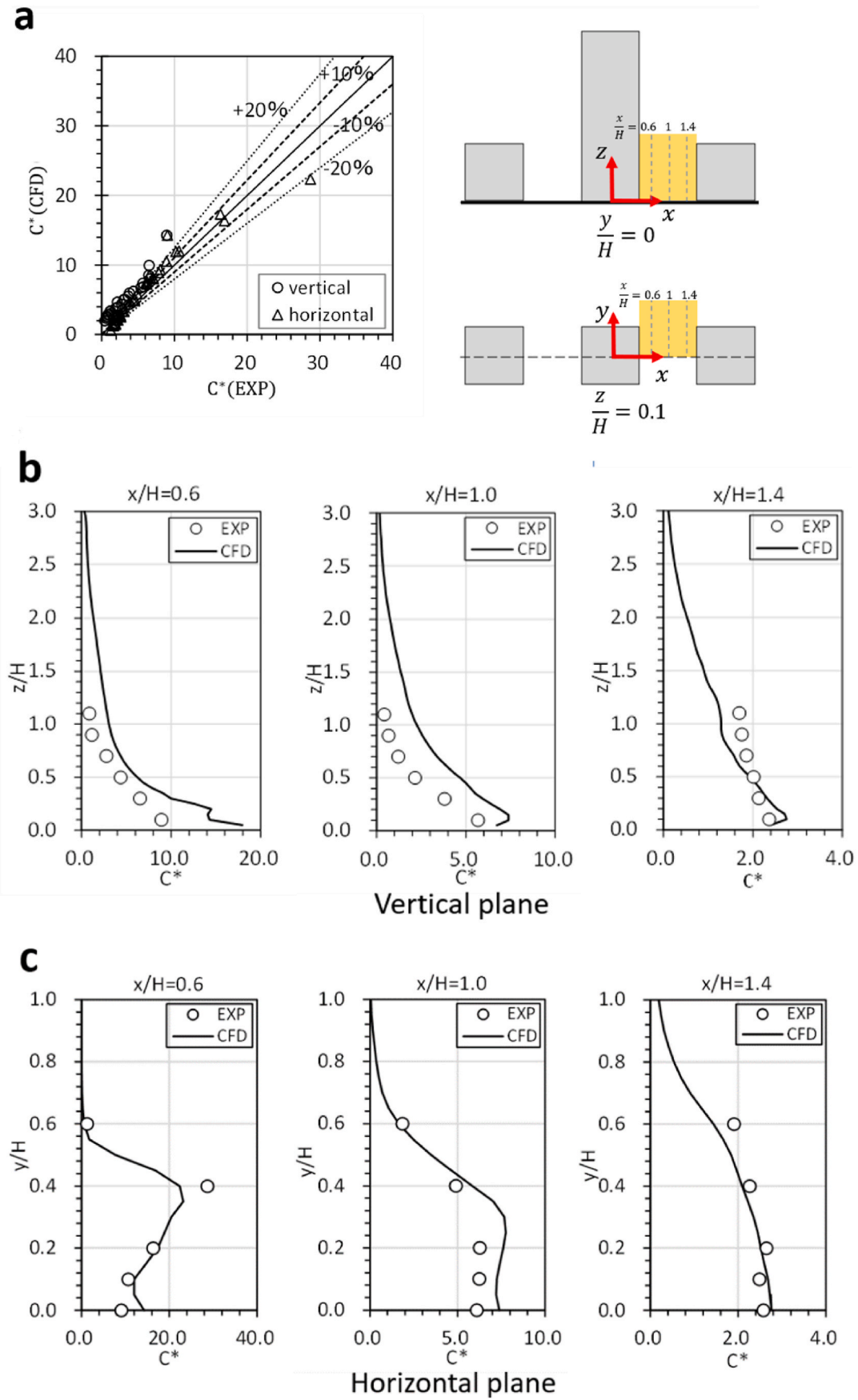


Fig. 6. (a) Scatter plot of the time-averaged concentration C^* over all measurement points on the vertical ($\frac{z}{H} = 0.1$) and horizontal ($\frac{y}{H} = 0$) planes. Vertical profiles (b) and horizontal profiles (c) of C^* over the selected lines by LES and the experiment for the surrounded case.

are also plotted in this figure. There is a good agreement between the predicted values by LES and those measured in the experiment for most of the measurement points. There are very few measurement points with absolute deviations of more than 20% at high-concentration regions. The vertical profiles of C^* over three vertical lines at $\frac{x}{H} = 0.6$, $\frac{x}{H} = 1.0$, and $\frac{x}{H} = 1.4$ is shown in Fig. 6(b). For all the vertical lines, the maximum C^* was measured near the ground level and it decreased uniformly along the z direction due to the mixing with the fresh air. The maximum deviations were observed over the vertical line at $\frac{x}{H} = 0.6$, which is close to the pollutant source on the leeward of the central high-rise building. This deviation may be attributed to the uncertainty of experimental measurement for the positioning of the sensor head of the carbon analyzer in the vicinity of the pollutant source in the downwind cavity and the flow disturbance due to the sensor support. The horizontal profiles of C^* over the same selected lines are shown in Fig. 6(c), that also show the good agreement between the LES predictions and the experimental measurements. Over the horizontal line at $\frac{x}{H} = 0.6$, C^* increased from about 10 at $\frac{y}{H} = 0$ to 30 at $\frac{y}{H} = 0.4$ near the corner edge of the central high-rise building, which is due to the pollutant trapped in the wake behind the building. C^* decayed rapidly along the y direction due to the incoming high velocity fresh air in the street canyon. More details about the mixing of pollutants and fresh air will be discussed in the following sections.

It should be noted that despite its importance, validation of the prediction accuracy of concentration fluctuations has rarely been done in previous studies. The scatter plot of the normalized root mean square (RMS) of concentration c_{rms}^* is shown in Fig. 7(a) over the measurement points on the horizontal and vertical planes. The overall accuracy of LES was reasonable with many points within the 10% and 20% deviation lines. However, there are also measurement points with deviations more than 20% in the range of $10 \leq c_{rms}^* \leq 20$ where LES showed over-prediction. The specific reason for this deviation is not clear. The vertical and horizontal profiles of c_{rms}^* over the selected lines are shown in Fig. 7 (b) and (c), respectively. The vertical profiles of c_{rms}^* shows almost the same trend as it was observed for C^* in Fig. 6(b) with high values of c_{rms}^* near the ground and near the pollutant source while it decayed rapidly along the z direction. The horizontal profiles of c_{rms}^* , as depicted in Fig. 7 (c), shows almost similar trend as C^* where it increased slightly along the y direction and decreased sharply in the street canyon.

The validation metrics for C^* and c_{rms}^* is shown in Table 3 assuming $D = 0.25$ and $W = 0.005$. For the time-averaged concentration, the calculated validation metrics are $q = 0.57$, $FAC2 = 0.83$, $FB = -0.19$, and $NMSE = 0.12$, which are all in the acceptable range except the hit rate q . The validation metrics for c_{rms}^* were close to the acceptable range for $FAC2$, FB , and $NMSE$, with values being calculated as 0.95, -0.34 , and 0.46, respectively. The hit rate for c_{rms}^* is quite low with a value of 0.33, which can be attributed to the deviations of LES and experimental results over the measurement points near the pollutant source on the leeward façade. The same large deviations were also reposted in Ref. [65] for an isolated cube. It is worth noting that the hit rate strongly depends on D and W , which represent acceptable relative and absolute differences in the comparison data. For $D = 0.35$ and $W = 0.05$, the hit rate for C^* and c_{rms}^* increased to 0.71 and 0.69, respectively, which are noticeably higher than the values shown in Table 3. Considering the higher measurement uncertainties of concentration measurement in comparison with the velocity measurements and also the noticeable variation of concentration measurement uncertainties at different measurement points, it is reasonable to assume larger values of D and W than those considered for the velocity measurements.

5. Impacts of surrounding buildings on pollutant dispersion

5.1. Velocity field

In the following, the main features of the turbulent flow field around

the isolated and surrounded cases are discussed to find out the effect of the surrounding buildings on pollutant dispersion. The streamlines and the contours of the normalized time-averaged velocity magnitude (V_m/U_H) are displayed over the vertical and horizontal planes in Fig. 8 for the isolated and surrounded cases. For the isolated case, as shown in Fig. 8(a), there is a strong downwash formed around the stagnation area at $\frac{z}{H} = 2.2$, while an upward flow was generated toward the building roof. The interaction of this upward flow and the incoming high-speed flow resulted in flow separation and reattachment over the roof area. A vertically long wake region was formed behind the building which features a strong upward flow over the leeward façade followed by a reattachment on the ground approximately at $\frac{x}{H} = 1.8$. A strong recirculating flow was formed in the vicinity of the roof trailing edge because of the interaction between the uprising flow and rooftop flow. For the surrounded case, as shown in Fig. 8(b), the downwash over the windward façade was accelerated by the roof flow of the upwind low-rise building resulting in the formation of a high-speed region over the building façade at $\frac{z}{H} = 0.8$ and a recirculating flow in the upwind cavity with quite high velocity magnitude of up to $\frac{V_m}{U_H} = 1$. The roof top flow was almost similar for both cases while the wake flow behind the high-rise building was interrupted by the downwind low-rise building. As a result, the velocity magnitude in the downwind cavity and near the leeward façade of the high-rise building was generally larger than those observed for the isolated case.

As depicted in Fig. 8(c), the mean flow field of the isolated case over the horizontal plane shows a flow diversion toward the lateral sides in front of the windward façade of the high-rise building which interacted with the free stream flow and formed a reverse flow and weak recirculation on the side walls. Two leeward wakes were also formed behind the building (only half of the domain is shown due to the symmetric distribution of the time-averaged flow over the horizontal plane). In comparison, as shown in Fig. 8(d), the mean flow distribution is much more complicated for the surrounded case due to the presence of the surrounding low-rise building. Only the main flow features around the central high-rise building are compared with those observed for the isolated case in the following. The strong downwash over the windward façade of the high-rise building (showed in Fig. 8(b)), created a high-speed reverse flow over the horizontal plane in the upwind cavity with $\frac{V_m}{U_H}$ up to 0.7 which is noticeably higher than those found for the isolated case in the same region. The high-speed reverse flow in the downwind cavity was diverged toward the lateral direction and interacted with the high-speed street canyon flow and diverted it toward the cavity of the lateral low-rise buildings. The side wall flow for the surrounded case also showed higher velocity than the isolated case specifically near the leeward corner followed by a stronger wake flow in the leeward cavity behind the high-rise building. For the surrounded case in the center of the downwind cavity at $\frac{x}{H} = 1$ and $\frac{z}{H} = 0.1$, the time-averaged velocity magnitude is significantly higher than the one predicted for the isolated case with values up to $\frac{V_m}{U_H} = 0.41$ whereas it was around 0.2 for the isolated case.

The contours of the normalized TKE (k/U_H^2) over the vertical and horizontal planes are compared in Fig. 9 for the isolated and surrounded cases. Over the vertical plane for the isolated case, shown in Fig. 9(a), the typical distribution of TKE around a bluff body is observed featuring large TKE values over the rooftop and in the wake behind the building. For the surrounded case, shown in Fig. 9(b), the TKE level on the rooftop and in the wake region behind the high-rise building above the street canyon ($\frac{z}{H} \geq 1$) is almost the same as the one calculated for the isolated case. The most notable difference in the TKE field between the isolated and surrounded cases was found near the leeward façade of the high-rise building below the canyon height ($\frac{z}{H} \leq 1$). As shown in Fig. 9(b), due to the strong reverse flow near the ground in the downwind cavity of the surrounded case, the TKE level is much higher than the one shown for the isolated cases in Fig. 9(a), which is almost four times larger. The TKE

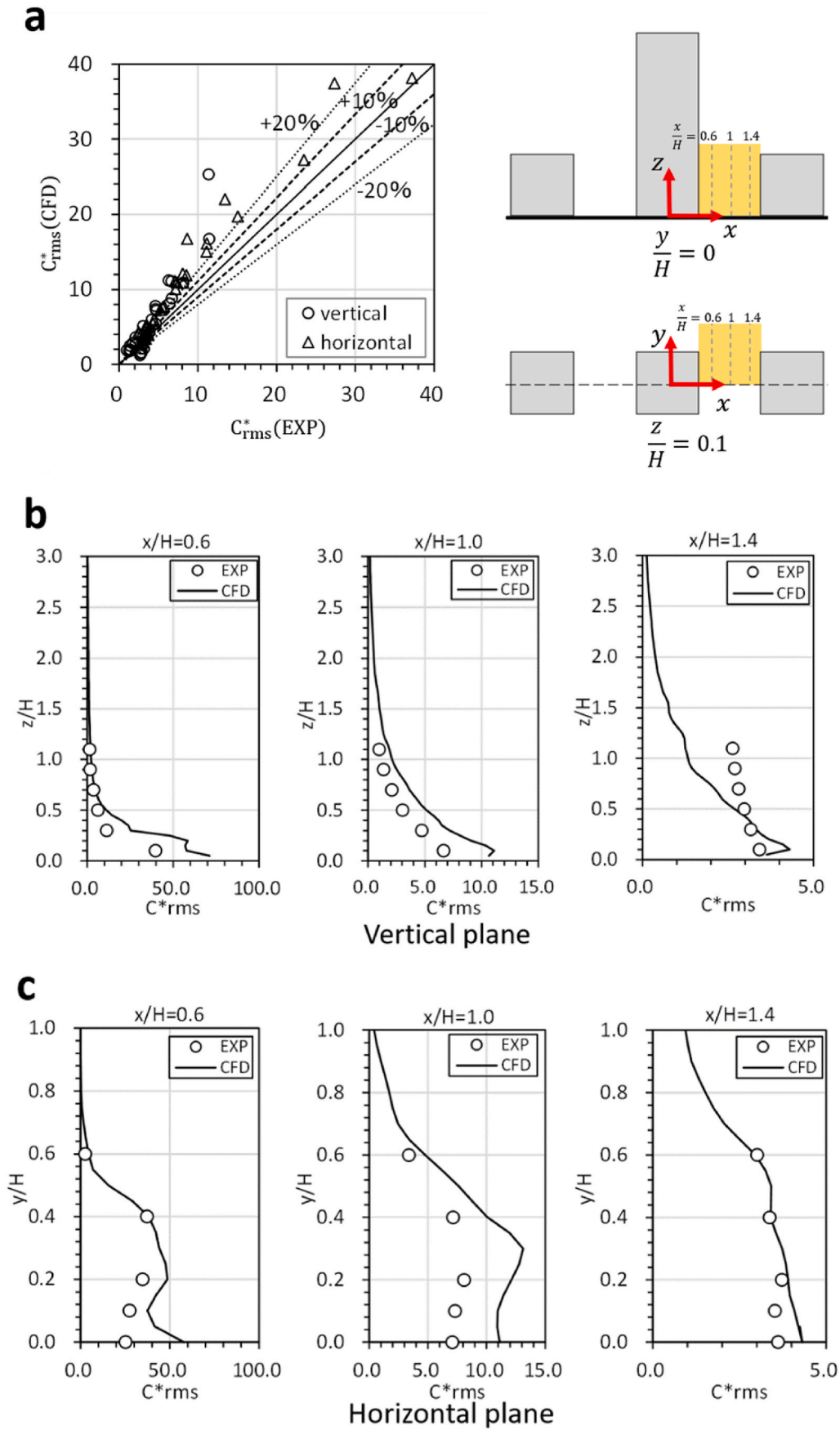


Fig. 7. (a) Scatter plot of the normalized rms of concentration C_{rms}^* over all measurement points on the vertical ($\frac{z}{H}=0.1$) and horizontal ($\frac{y}{H}=0$) planes. Vertical profiles (b) and horizontal profiles (c) of C_{rms}^* over the selected lines by LES and the experiment for the surrounded case.

Table 3

Validation metrics for the normalized time-averaged concentration C^* and its rms c_{rms}^* for the surrounded building.

	Hit rate (q)	FAC2	FB	NMSE
C^*	0.576	0.833	-0.198	0.127
c_{rms}^*	0.333	0.955	-0.345	0.465

level around the windward façade of the high-rise building was also higher in the surrounded case than the isolated case, however, this difference probably has less contribution to the concentration field as the pollutant source is on the leeward façade.

Over the horizontal plane for the isolated case, as displayed in Fig. 9 (c), the TKE peaked around the side wall and in the wake behind the building with a peak value of about $\frac{k}{U_H^2} = 0.15$. This significant TKE peak in the wake behind the building was mainly generated by a large-scale fluctuation due to vortex shedding. The surrounded case, in contrast, showed a lower TKE level around the sidewall while the TKE increased to values of about $\frac{k}{U_H^2} = 0.15$ along the interaction line between the

reverse flow from the upwind cavity and the incoming street canyon flow (see Figs. 8(d) and Fig. 9(d)). The higher TKE level inside the downwind cavity and near the leeward façade near the pollutant source of the surrounded case is clearly shown in Fig. 9(d).

5.2. Concentration field

In Fig. 10(a), the contours of normalized time-averaged concentration C^* over the vertical plane is shown for the isolated case. A logarithmic scale was used in the color bar for better visualization. The maximum concentration was predicted in the vicinity of the pollutant source over the leeward façade with values of about $C^* = 45$ that decayed with different rates in the streamwise and vertical directions. For the surrounded case, as depicted in Fig. 10(b), the pollutant concentration accumulated in the downwind cavity resulted in noticeably higher C^* , in comparison with the isolated case. For instance, at the pedestrian height of $\frac{z}{H} = 0.1$ at $\frac{x}{H} = 1.0$ (Point A in Fig. 10), C^* is 3.2 and 7.5 for the isolated case and surrounded case, respectively which means an increasing ratio of about 2 for the surrounded case. This highlights

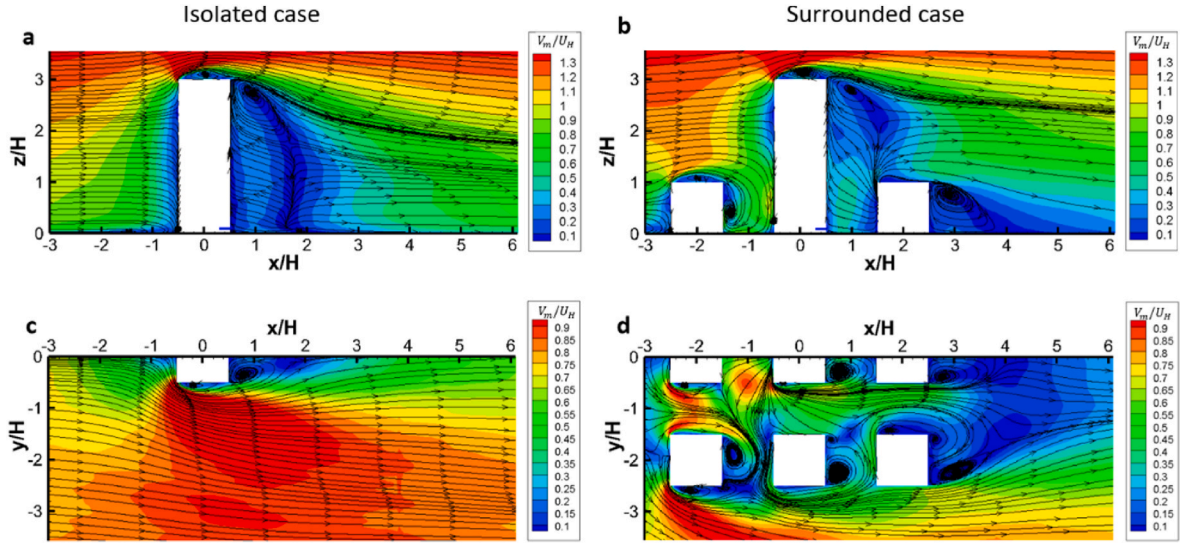


Fig. 8. Streamlines of the normalized time-averaged velocity magnitude V_m/U_H for the isolated and surrounded buildings over the horizontal ($\frac{z}{H} = 0.1$) and vertical ($\frac{y}{H} = 0$) planes.

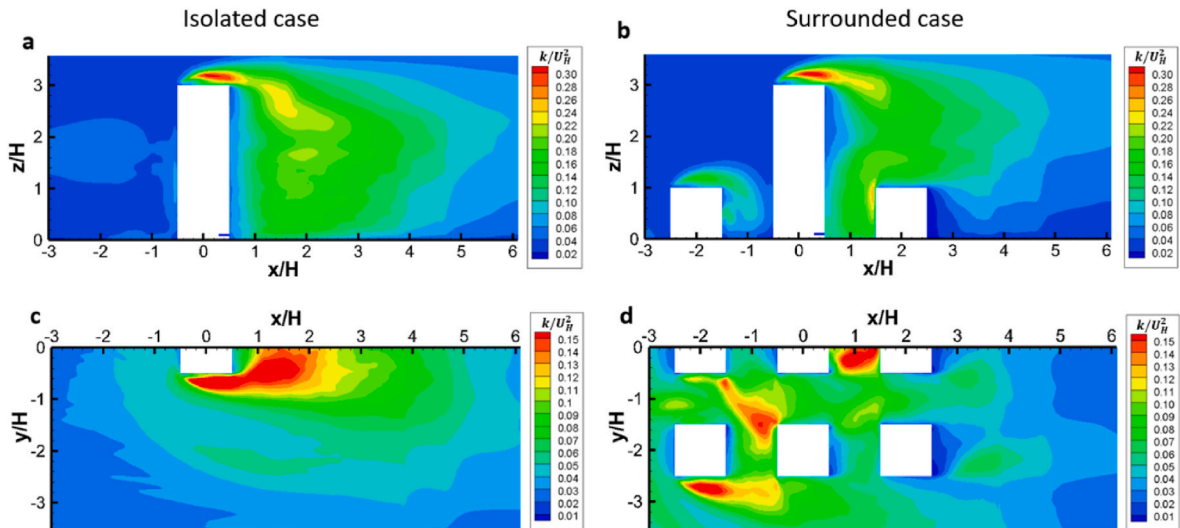


Fig. 9. Contours of normalized turbulent kinetic energy k/U_H^2 for the isolated and surrounded buildings over the horizontal ($\frac{z}{H} = 0.1$) and vertical ($\frac{y}{H} = 0$) planes.

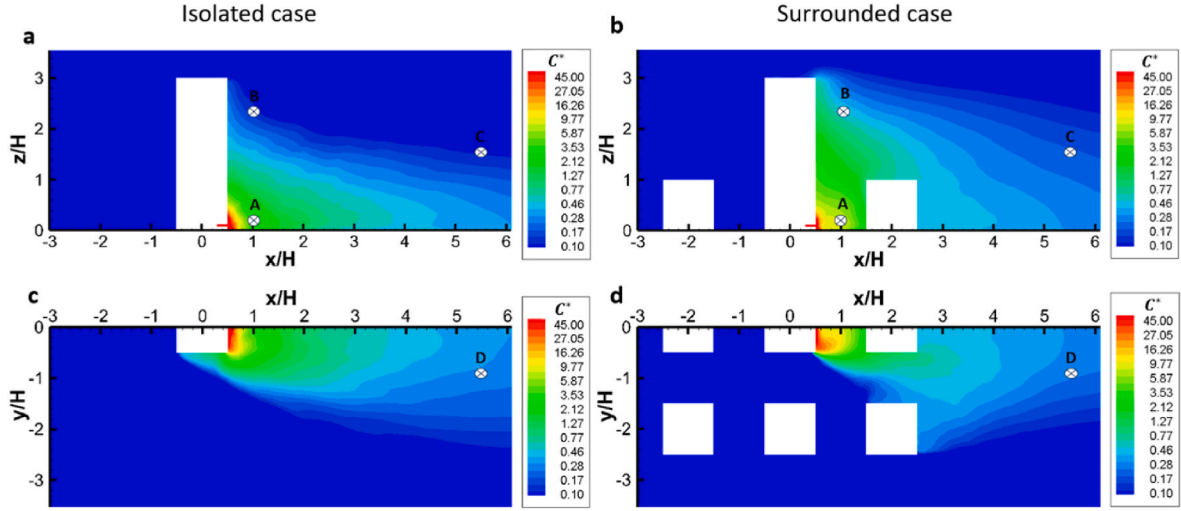


Fig. 10. Contours of normalized time-averaged concentration C^* for the isolated and surrounded building cases over the horizontal ($\frac{z}{H} = 0.1$) and vertical ($\frac{y}{H} = 0$) planes.

the influence of surrounding buildings on the mean concentration field. Another important difference between the two cases is observed in the higher distribution of C^* in the wake region above the canyon height for the surrounded case. For example, at a height of $\frac{z}{H} = 2.5$ at $\frac{x}{H} = 1$ (Point B in Fig. 10), the time-averaged concentration for the surrounded case is 5 times larger than the isolated case, which means the residents on higher floors experience more pollutant concentration in the surrounded case even though they are located much higher than the canyon rooftop. Another interesting difference between the distribution of pollutant concentration for the isolated and surrounded cases can be seen in the far field behind the high-rise building. For the isolated case, C^* was stretched in the streamwise direction, whereas C^* was expanded not only in the streamwise direction but also in the vertical direction for the surrounded case. At a far point in $\frac{x}{H} = 5.5$ and $\frac{z}{H} = 1.5$ (Point C in Fig. 10), $C^* = 0.08$ for the isolated case while for the surrounded case $C^* = 0.18$. This fact suggests that there is a risk of underestimating pollutant concentrations around a high-rise building, especially near the upper floors if the surrounding buildings were not properly considered.

For a better 3D visualization of the difference in the pollutant dispersion between the isolated and surrounded cases, the iso-surface of C^* is shown for $C^* = 1.86$ and 0.28 in Fig. 11 for both cases. The high-concentration iso-surface of $C^* = 1.86$, which is shown by the red color, was formed around the pollutant source behind the building. This area is noticeably smaller in the isolated case than the surrounded case in which the high-concentration surface was extended vertically to a height of about $2/3H$ and almost covered all the downwind cavities. The iso-surface of low-concentration ($C^* = 0.28$), which is shown by light blue color, covered a large area not only in the streamwise direction but also extended over the side walls of the high-rise building due to the large-scale fluctuation of the flow recirculation near the leeward corner

and over the side wall (see Fig. 8(c)) which redirected the pollutant from the leeward cavity to the side wall region. In contrast, for the surrounded case, the high-velocity street canyon flow around the side walls of the high-rise building (see Fig. 8(d)), preventing the suction of pollutant from the downwind cavity to the side wall. However, the pollutant was diverted vertically due to the strong upward flow over the leeward façade of the high-rise building (see Fig. 8(d)) for the surrounded case and then it was extended to the side walls and roof edge. The pollution also penetrated well in the wake region of the downwind buildings in both the streamwise and lateral directions due to the complex flow interaction in the wake region which features vertically long recirculation behind the side low-rise building (see Fig. 8(d)).

5.3. Pollutant transport mechanism

The total mass flux in direction i can be decomposed into two parts [66]:

$$\overline{u_i c} = U_i \bullet C + \overline{u_i' c'} \quad (6)$$

The first term on the right-hand side is the mean convective mass flux (CF_i) representing the contribution of the mean flow and the second term is the turbulent diffusive mass flux (TF_i) representing the contribution of the turbulent fluctuation. The instantaneous total mass flux was sampled and averaged for a sampling time of 45 s to calculate $\overline{u_i c}$. CF_i was calculated directly by multiplying the time-averaged velocity U_i and time-averaged concentration C while TF_i was calculated by subtracting CF_i from $\overline{u_i c}$. In this analysis, the SGS components of the fluxes are ignored because their contributions are expected to be small as described in Section 4.1.

The contours of the convective mass flux in the streamwise (CF_x),

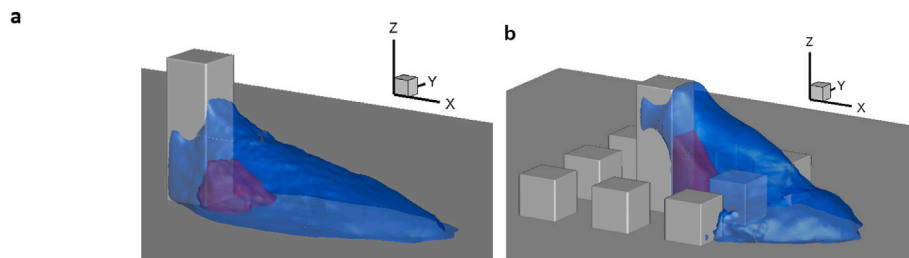


Fig. 11. Iso-surface of time-averaged concentration for $C^* = 1.86$ (red surface) and $C^* = 0.28$ (light blue surface) for the isolated (a) and surrounded (b) cases. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

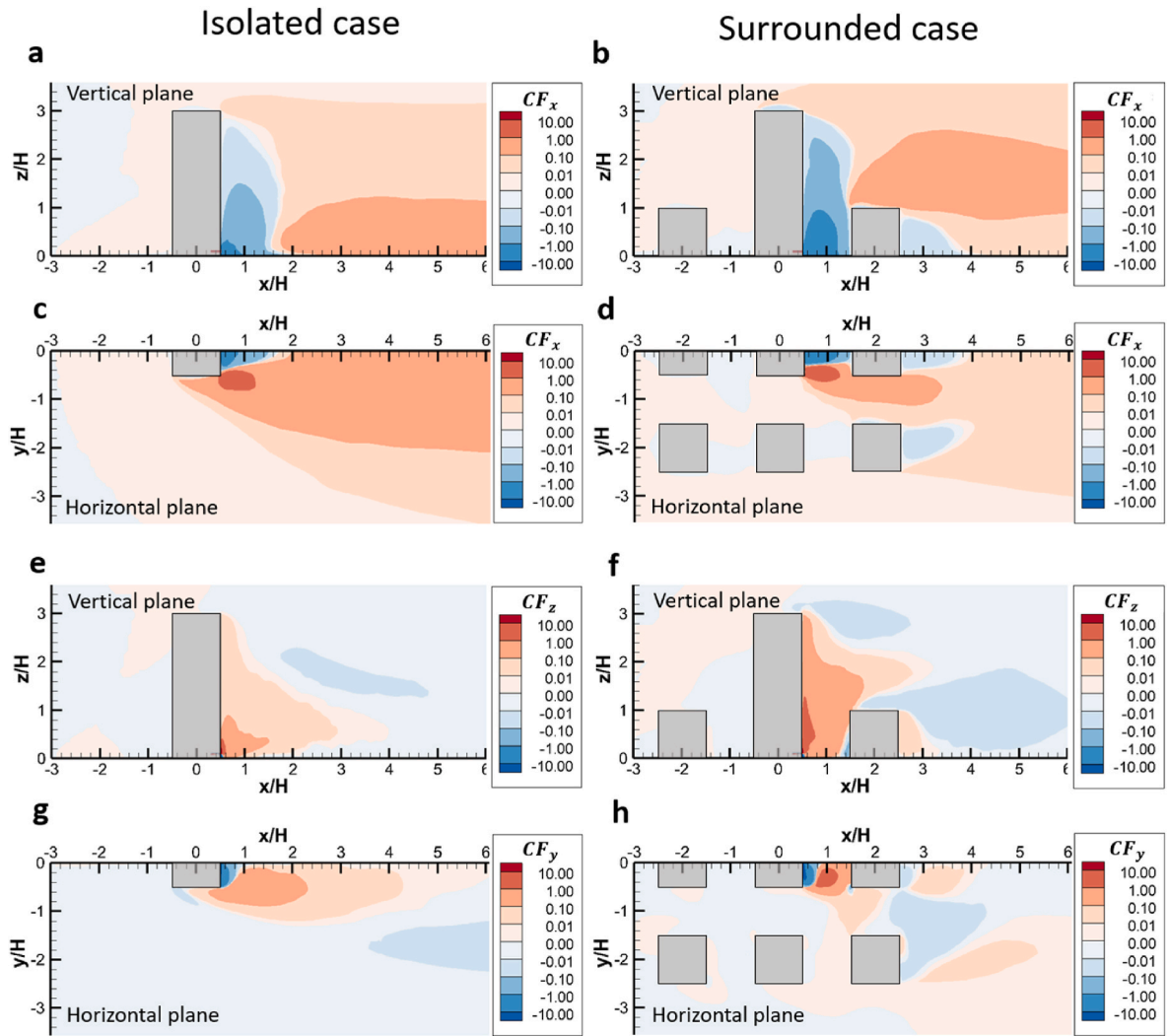


Fig. 12. Contours of the convective fluxes for the isolated and surrounded cases over the horizontal and vertical planes.

lateral (CF_y), and vertical (CF_z) directions are depicted in Fig. 12 over the vertical and horizontal planes for the isolated and surrounded cases. Over the vertical plane for the isolated case, as shown in Fig. 12(a), a large area with negative streamwise convective mass flux CF_x was formed behind the high-rise building with $-10.0 \leq CF_x \leq 0.0$ which is due to the reverse flow in this region (see the streamlines in Fig. 8(a)) which transfers the pollutant in the reverse direction against the main wind direction. The maximum positive streamwise convective mass flux was observed near the ground level after the boundary layer reattachment with values in the range of $0.1 \leq CF_x \leq 1.0$. This region is responsible for the washing away of the pollutant and reducing the mean concentration as shown in Fig. 10(a). For the surrounded case, as seen in Fig. 12(b), the contribution of the negative convective mass flux is higher than the isolated case in the downwind cavity due to the stronger reverse flow enhanced by the downstream low-rise building. Another difference can be seen in the location of the positive streamwise mass flux for the surrounded case which was formed above the canyon height ($z/H \geq 1$). The negative streamwise mass flux has a negative side effect from the internal air quality point of view because it shows that the mass flux may penetrate the central high-rise building through the openings on the leeward façade. This phenomenon is more challenging for the surrounded case as the negative streamwise mass flux covered a larger area and extended vertically toward the building roof. Over the horizontal plane for the isolated case (Fig. 12(c)), it can be seen that the convective mass flux was positive around the side wall while, as shown

in Figs. 10(a) and Fig. 11, the pollutant was redirected there in the reverse direction to the convective mass flux. This interesting finding highlights the contribution of the turbulent mass flux, which will be discussed in the following. In contrast, the distribution of the time-averaged concentration (see Fig. 10(b)) for the surrounded case was compatible with the mean streamwise convective mass flux as shown in Fig. 12(d). In this case, the location of the regions with negative streamwise convective mass flux corresponds to the time-averaged streamlines.

Fig. 12(e) displays the contours of the vertical convective mass flux CF_z for the isolated case and highlights the mean flow contribution over the leeward façade of the high-rise building where the mean vertical flow extracts the pollutant toward the roof height. For the surrounded case, the vertical convective flux in the downwind cavity and along the leeward façade of the high-rise building was noticeably larger than the isolated case with $CF_z \approx 1.0$ for the surrounded case in comparison $CF_z \approx 0.1$ for the isolated case. The contribution of the lateral convective mass flux (CF_y), which is shown on the horizontal plane in Fig. 12(g and h), showed the importance of the lateral mass transport of the pollutant dispersion behind the isolated and surrounded cases. Near the pollutant source, there was a region with a large negative lateral convective mass flux with $CF_y \approx -10.0$ while the flux sign changed rapidly to positive values with $CF_y \approx +1.0$ for the surrounded case. The contribution of the positive lateral convective mass flux in the pollutant transfer inside the downwind cavity for the surrounded case is

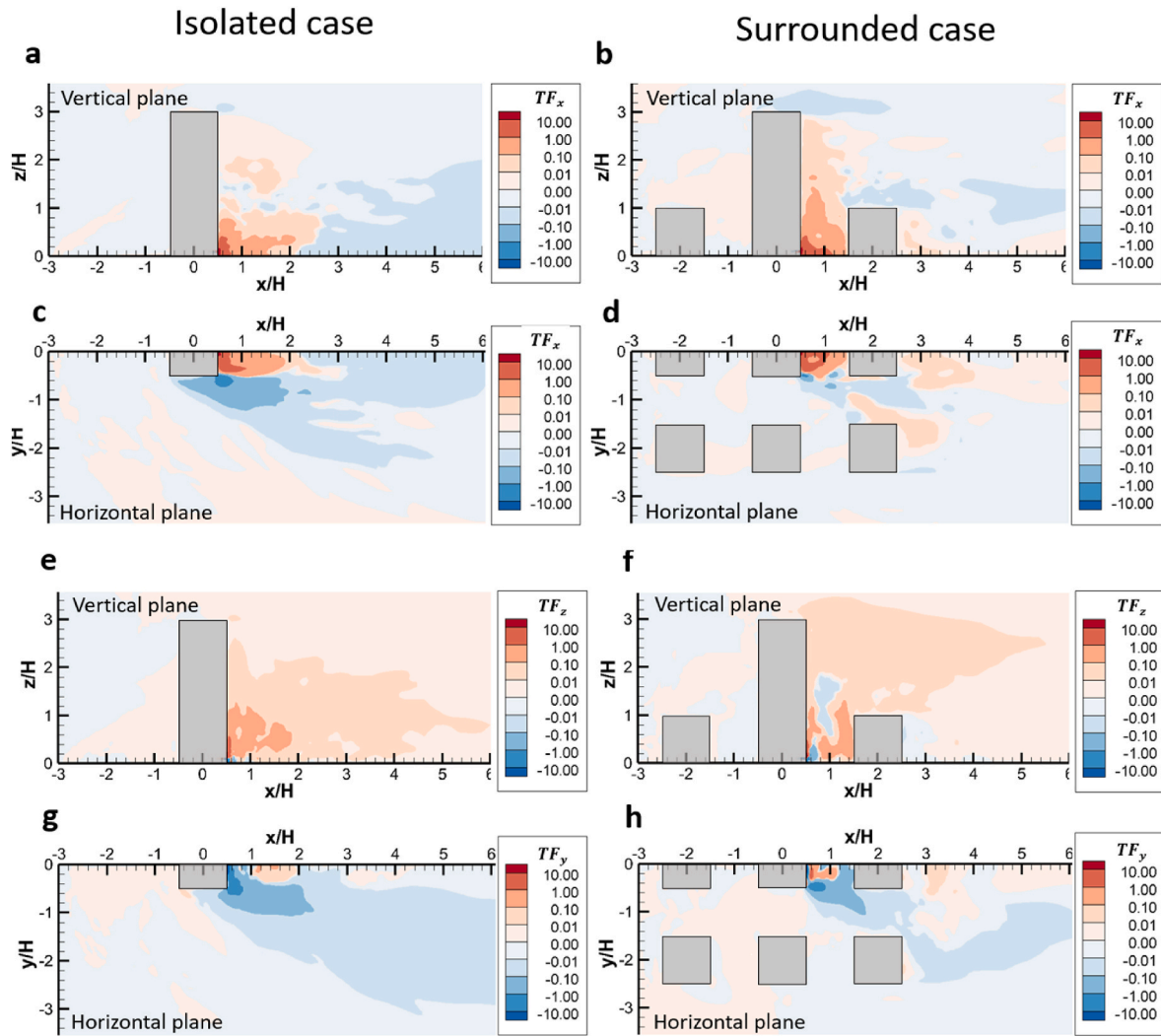


Fig. 13. Contours of the diffusive turbulent fluxes for the isolated and surrounded cases over the horizontal and vertical planes.

noticeably higher than the isolated case where there were two regions with $CF_y \cong 10.0$ and -10.0 .

The contours of the turbulent diffusive mass flux are shown in Fig. 13 over the same vertical and horizontal planes shown in the previous figure. For a better understanding of the contribution of the turbulent mixing on the pollutant transfer, it is useful to compare the direction of the time-averaged flow shown in Fig. 8, the direction (sign) of convective mass flux shown in Fig. 12, and the direction (sign) of the turbulent mass fluxes shown in Fig. 13. For the isolated case, as shown in Fig. 13(a) over the vertical plane, interestingly the direction of the turbulent mass flux in the streamwise direction ($TF_x = \overline{u'c}$) is positive with values between 0.1 and 10 near the pollutant source and the ground behind the high-rise building. It highlights the high contribution of turbulent mixing and it also shows that turbulent mixing competes with the convective mass flux (mean flow mixing) because they have different signs. The same tendency was observed for the surrounded case as shown in Fig. 13(b) where despite a strong reverse flow in the downwind cavity and negative streamwise convective mass flux the streamwise turbulent mass flux was positive (see Figs. 7(b) and Fig. 12(b)). For the surrounded case, the contribution of turbulent mixing in the streamwise direction was larger than the isolated case in the vicinity of the leeward façade. Over the horizontal plane of the isolated case, as shown in Fig. 13(c), a large area of negative streamwise turbulent mass flux was formed around the side wall of the high-rise building. It is responsible for the increased concentration in this region (see Fig. 10(a) while the mean

flow and convective mass flux are positive and have opposite signs. It can be related to the high TKE level around the side wall of the isolated case as shown in Fig. 9(a) which was due to a large fluctuating streamwise velocity that was measured experimentally and explained in detail in the previous paper [67]. Over the horizontal plane of the surrounded case, as shown in Fig. 13(d), the contribution of the turbulent mixing in the streamwise direction was much smaller than the isolated case over the side wall of the high-rise building. In this region, as displayed in Fig. 8(b), the high-speed street canyon flow prevents the turbulent mixing and pushes the pollutant away in the streamwise direction in contrast to the isolated case.

The vertical turbulent mass flux ($TF_z = \overline{w'c}$) was positive over the vertical plane behind the isolated high-rise building with values between 0.1 (far from the pollutant source) and 10 (near the pollutant source) as shown in Fig. 12(e). For the surrounded case over the vertical plane, as shown in Fig. 12(f), the distribution of TF_z was more complicated and non-uniform in comparison to the isolated case which is due to the complex flow pattern in the downwind cavity. The vertical convective mass flux was more uniform, and it was dominant for the mixing of the pollutant in the downwind cavity. In comparison with the isolated case, the region of large positive vertical turbulent mass flux was extended vertically even to the roof height and then stretched in the streamwise direction. This highlights the importance of considering the effect of the surrounding buildings on the turbulent mixing over the leeward façade of the high-rise building which can be important for

indoor air quality evaluation as well as outdoor air quality prediction. The distribution of the lateral turbulent mass flux ($TF_y = \overline{v'c'}$) for the isolated case shows the formation of a large area with a negative sign around the side wall which was extended noticeably in the streamwise and lateral direction as seen in Fig. 12(g). This was more complicated for the surrounded case where two regions of large TF_y with opposite signs formed in the downwind cavity that showed a significant contribution to the turbulent mixing in the lateral direction, as depicted in Fig. 13(h).

For a better evaluation of the influence of the surrounding buildings on the concentration distribution on volumetric bases, the pollutant mass transfer rate budget was calculated for three different control volumes (CV) around the central high-rise building as shown in Fig. 14. The dimension of all CVs was equal to $H \times H \times H$. The mean convection and turbulent components of the pollutant mass transfer rate (m_p) were calculated over each surface of each CV by multiplying the convective and turbulent diffusive fluxes by the cell area (A_j) and density (ρ_j):

$$m_{p_conv} = \sum_{j=1}^m U_{jn} C_j \rho_j A_j \quad (7)$$

$$m_{p_turb} = \sum_{j=1}^m \overline{u'_{jn} c_j} \rho_j A_j \quad (8)$$

Here u_{jn} and c_j are the instantaneous normal velocity and concentration in cell element j , and m is the total number of cell elements over each surface. Depending on the surface direction, m_{p_conv} and m_{p_turb} were separated into incoming and outgoing components with (+) and (−) sign. The positive flux means the flux entered the CV while the negative flux means that the flux exited the CV through the corresponding surface. Because both incoming and outgoing mass transfers can occur even on a single CV surface by location, both signs of the flux may appear on a surface. The pollutant mass transfer rates were normalized by the total pollutant mass flow rate injected through the source (m_{p_total}), defined as

$$m_{p_conv}^* = \frac{m_{p_conv}}{m_{p_total}} \text{ and } m_{p_turb}^* = \frac{m_{p_turb}}{m_{p_total}}$$

The pollutant mass transfer rate budgets for upwind (UP), downwind (DW), right (RS), left (LS), and top (TP) surfaces of each CV are compared in Fig. 14 for the isolated and surrounded cases. The surfaces blocked by the building walls, where the fluxes cannot be defined, are indicated as 'x' on the horizontal axis. For CV1, which is located beside the high-rise building, $m_{p_conv}^*$ and $m_{p_turb}^*$ were almost zero for UW and

RS for the isolated case, as shown in Fig. 14(a). On the DW surface, the total transfer rate is almost zero, because the incoming convective mass transfer rate was $m_{p_conv}^* = +30\%$ while the outgoing turbulent mass transfer rate was $m_{p_turb}^* = -30\%$. This shows that the pollutant accumulation in CV1 was obtained as a competition between the turbulent and convective mass fluxes which carried the pollutant inside and outside of the domain respectively. This is also compatible with the contours of convective and turbulent mass fluxes around the side wall of the isolated case which had opposite directions (signs), as shown in Figs. 12(a) and Fig. 13(a). For the surrounded case, the mass transfer rate budget over the RS surface was near zero while for the DW surface, very small mass transfer rates were calculated as $m_{p_turb}^* = +1\%$ and $m_{p_conv}^* = -1\%$. For both cases, the mass transfer rate budget on TP surface was near zero.

For CV2, which is located behind the central high-rise building, the mass transfer rate budgets are illustrated in Fig. 14(b). For the isolated case on the DW surface, the incoming convective mass transfer rate was $m_{p_conv}^* = +2\%$ while the outgoing convective and turbulent mass transfer rates were noticeably large with $m_{p_mean}^* = -14\%$ and $m_{p_turb}^* = -11\%$. For the isolated case, most of the pollutant was carried away from CV2 by the mean and turbulent mass flow rates through the LS and RS surfaces with $m_{p_conv}^* = -8\%$ and $m_{p_turb}^* = -42\%$. The pollutant was also drained through the TP surface by convective and turbulent mass transfer rates with values of $m_{p_conv}^* = -7\%$ and $m_{p_turb}^* = -9\%$. For the surrounded case, as the DW and UW surfaces are blocked by the building walls, the majority of the pollutant was drained by turbulent mixing through RS and LS surfaces ($m_{p_turb}^* = -55\%$) and convection through the TP surface ($m_{p_conv}^* = -59\%$). In CV2, the convection mass flux equally contributes on each surface for the isolated case, but in the surrounded case, the convection flux is dominant only on the upper surface due to the strong circulating flow behind the high-rise building.

The pollutant mass transfer rate budget for CV3, which was in the street canyon downstream of the central high-rise building, is depicted in Fig. 14(c). For the isolated case, most of the pollutants entered CV3 by turbulent mixing through the LS surface with $m_{p_turb}^* = 42\%$. The pollutant was extracted from CV3 by the mean flow through the DW and LS surfaces at a mass transfer rate of $m_{p_conv}^* = -37\%$ and $m_{p_conv}^* = -21\%$ for the isolated case. For the surrounded case, the pollutants enter the CV by turbulent mixing through the LS surface with $m_{p_turb}^* = 55\%$

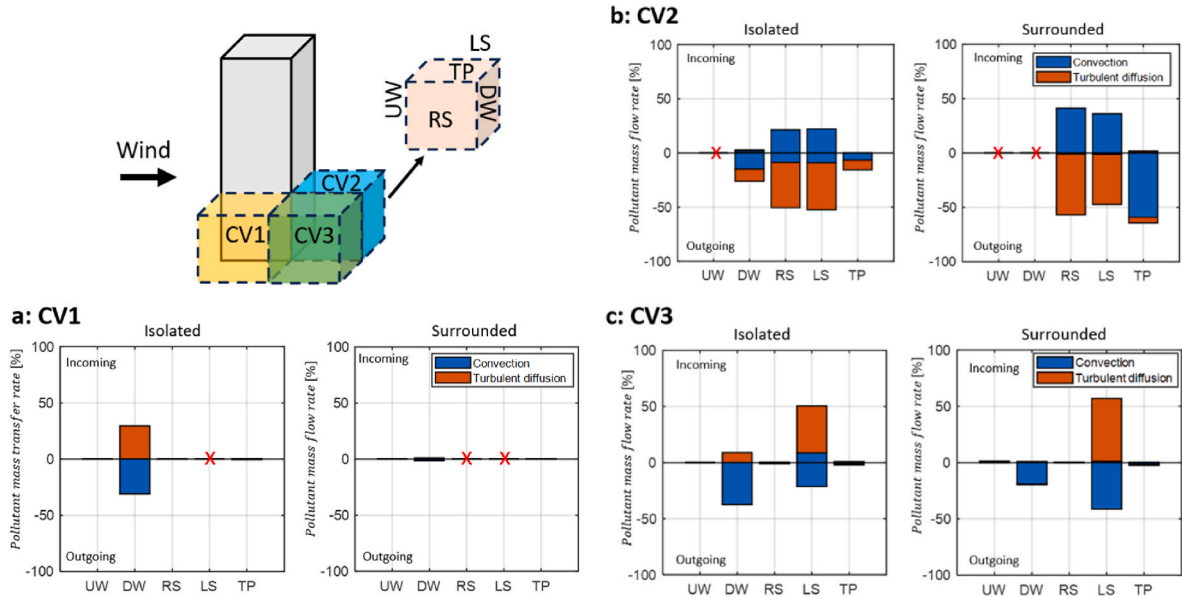


Fig. 14. Pollution mass flow rate budget for three different control volumes (CV) around the central high-rise building for the isolated and surrounded cases.

while it left the CV by mean flow through the same surface at a rate of $m_{conv}^* = -41\%$. In the surrounded case, mass transfer by the downstream convection (DW side), was almost half of the one calculated for the isolated case. Thus, it is demonstrated that the presence of surrounding buildings significantly alters the pollutant transfer mechanism around a high-rise building.

6. Conclusions

The pollutant dispersion around a high-rise building was studied numerically and experimentally for two cases with an isolated building and with several low-rise buildings surrounding the central high-rise building. The numerical model based on LES was validated and then utilized to investigate the effect of surrounding buildings on the pollution transfer mechanism around the high-rise building when the pollutant was released through a hole on the leeward façade of the high-rise building at the pedestrian height. In addition to the flow field parameters, e.g. mean velocity magnitude and TKE, the distributions of mean and rms values of the concentration, and the convective and turbulent diffusive fluxes were analysed. The following are the important outcomes of the study:

- The presence of surrounding buildings significantly alters the pollutant transfer mechanism around a high-rise building. Pollutants decayed in the streamwise and vertical directions with different rates for the isolated and surrounded cases.
- For the surrounded case, the pollutant accumulates in the downwind cavity with a mean concentration almost twice the isolated case. The contribution of the negative convective mass flux was higher than the isolated case in the downwind cavity due to the stronger reverse flow enhanced by the downstream low-rise building.
- In the surrounding case, the pollutant in the wake region shifted vertically toward the roof of the high-rise building. For a location at 80% of the building height (2.5H), the mean concentration was five times larger than the isolated case. It can be interpreted that the residents on higher floors experience more pollutant concentration than the isolated case even though they are located much higher than the canyon rooftop.
- Over the leeward façade of the high-rise building, the contribution of the turbulent mixing in the streamwise direction was noticeable. It also competes with the convective mass flux because they had opposite directions.
- On the side walls of the isolated building, the turbulent diffusive mass flux was negative and diverted the pollutant from the downwind cavity into the side wall area. In contrast, the strong streamwise mean flow in the street canyon of the surrounded case pushed the pollutant in the streamwise direction.
- For a control volume (CV) on the side of the building, the pollutant accumulation was obtained as a competition between the turbulent and convective mass fluxes which carried the pollutant inside and outside of the domain respectively.
- For the isolated case, most of the pollutant was carried away from downwind CV by turbulent mass transfer rates through the left and right side surfaces with $m_{p-turb}^* = -42\%$ which was 5 times larger than the convective mass transfer rate.
- For the surrounded case, the majority of the pollutant in the downwind CV was drained by turbulent mixing through right and left side surfaces ($m_{p-turb}^* = -55\%$) and convection through the top side surface ($m_{p-conv}^* = -59\%$).

In this study, only one wind direction was considered and the location of the pollutant was fixed. Although we consider this to be the most critical condition, to obtain a better picture of the pollution transfer mechanism, different wind angles and pollutant locations should also be considered in future work. Furthermore, the pollutant dispersion for

different flow regimes (e.g., isolated roughness flow, wake-interference flow, etc.) and building configurations will be considered in future works. Since only LES was considered for turbulence modeling in this study, a detailed comparison between different turbulence models will be considered in future works to analyze the limitation of RANS turbulence models for the prediction of pollutant dispersion in urban areas for practical applications.

CRedit authorship contribution statement

Mohammadreza Shirzadi: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation. **Yoshihide Tominaga:** Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgment

The authors thank Mr. Takahiro Suzuki, a technical staff member at the Wind and Fluid Engineering Research Center at the Niigata Institute of Technology, for his invaluable assistance with the wind tunnel experiment. The authors would also like to thank Ms. Fumiko Ibe, a member of the Wind and Fluid Engineering Research Center, for her significant contribution to the preparation of this manuscript.

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